

50.007 Machine Learning, Spring 2025
Homework 3

Due 11 April 2025, 11.59pm

In this homework, we would like to look at the Hidden Markov Model (HMM), one of the most influential models used for structured prediction in machine learning.

Question 1 [20 points]

Assume that we have the following training data available for us to estimate the model parameters:

State sequence	Observation sequence
(X, X, Z, X)	(b, c, a, b)
(X, Z, Y)	(a, b, a)
(Z, Y, X, Z, Y)	(b, c, a, b, d)
(Z, Z, Y)	(c, b, a)
(X, X)	(c, a)
(Z)	(d)

The HMM is defined by:

- The initial probabilities $\pi(s)$,
- The transition probabilities $a_{u,v} = P(v \mid u)$,
- The emission probabilities $b_u(o) = P(o \mid u)$.

(a) Clearly state the parameters associated with the HMM and, using maximum likelihood estimation (MLE), compute the optimal model parameters (i.e., the initial, transition, and emission probabilities). Assume that the state space is $\{X, Y, Z\}$ and the observation alphabet is $\{a, b, c, d\}$. Note that you should include the transitions from a special START state and to a STOP state as appropriate [10 points].

(b) Smoothing Extension. In practice, zero-counts can cause problems. Explain how you would apply Laplace smoothing to the MLE estimates obtained in part (a) for both transition and emission probabilities. Write down the smoothed probability formulas [10 points].

Question 2 [10 points]

Now, consider during the evaluation phase, you are given the following new observation sequence. Using the parameters (with unsmoothed MLE estimates) from Question 1, find the most probable state sequence using the Viterbi algorithm discussed in class. Clearly present the steps that lead to your final answer.

State sequence	Observation sequence
$(?, ?)$	(a, b)

Question 3 [20 points]

The HMM discussed in class makes a simple first-order assumption, where the next state only depends on the previous state in the generative process. However, it is possible to extend the model discussed in class to have second-order dependencies. In other words, the HMM can be parameterised in the following way:

$$p(x_1, \dots, x_n, y_1, y_2, \dots, y_n) = \prod_{i=1}^{n+1} p(y_i \mid y_{i-2}, y_{i-1}) \cdot \prod_{i=1}^n p(x_i \mid y_i),$$

where we define $y_{-1} = y_0 = \text{START}$ and $y_{n+1} = \text{STOP}$.

In other words, the transition probabilities are changed from $p(y_i \mid y_{i-1})$ to $p(y_i \mid y_{i-2}, y_{i-1})$. Describe the Viterbi algorithm used for decoding such a second-order HMM model. In other words, describe the dynamic programming algorithm that computes the following efficiently for such an HMM:

$$(y_1^*, y_2^*, \dots, y_n^*) = \arg \max_{y_1, y_2, \dots, y_n} p(x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n).$$

(a) Initialization and State Representation [4 points]

Describe how you would initialize the dynamic programming table for this second-order HMM. In your answer, define the state pair $\langle u, v \rangle$ (with u representing the state at time $i - 1$ and v at time i) and explain its significance in the algorithm.

(b) Recursion [8 points]

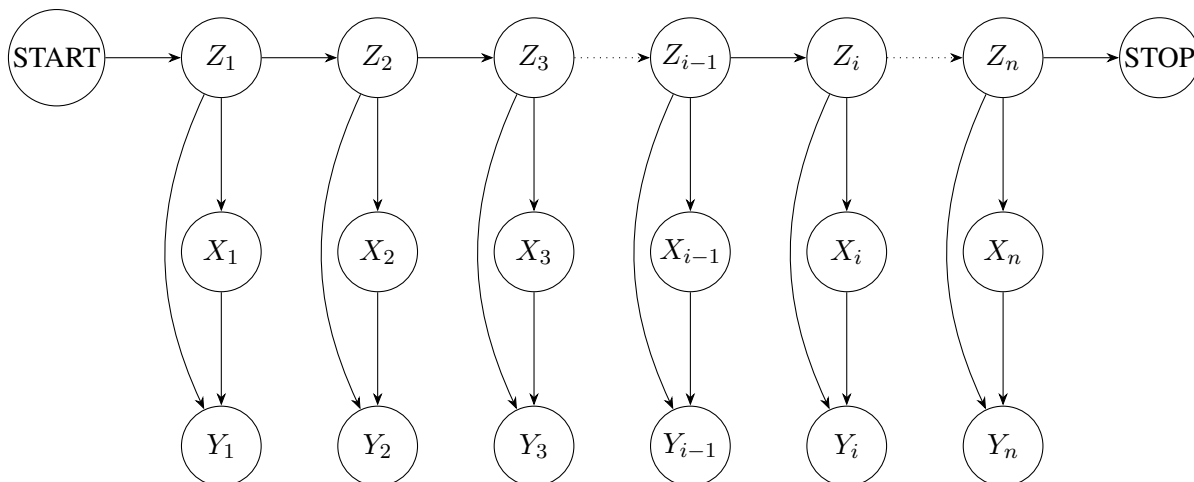
Derive the recurrence relation for the Viterbi algorithm in this second-order HMM. Show explicitly how the transition probabilities $p(y_i \mid y_{i-2}, y_{i-1})$ and the emission probabilities $p(x_i \mid y_i)$ are incorporated in the update step.

(c) Termination and Backtracking [8 points]

Explain the termination step (including the transition to the STOP state) and describe the backtracking procedure to recover the optimal state sequence $(y_1^*, y_2^*, \dots, y_n^*)$.

Question 4 [20 points]

Now consider a slightly different graphical model which extends the HMM (see below). For each state (Z), there is now an observation pair (X, Y) , where the Y sequence is generated from both the X sequence and Z sequence.



Assume you are given a large collection of observation pair sequences, and a predefined set of possible states $\{0, 1, \dots, N-1, N\}$, where $0 = \text{START}$ and $N = \text{STOP}$. You would like to estimate the most probable state sequence for each observation pair sequence using an algorithm similar to the dynamic programming algorithm discussed in class. Clearly define the forward and backward scores in a way analogous to HMM. Give algorithms for computing the forward and backward scores. Analyze the time complexity associated with your algorithms (for an observation pair sequence of length n).

1. a) $\pi(x) = \frac{3}{6} = \frac{1}{2}$ $\pi(y) = 0$ $\pi(z) = \frac{3}{6} = \frac{1}{2}$

Trans. probabilities:

u \ v	x	y	z	STOP
START	0.5	0	0.5	-
x	0.333	0	0.333	0.333
y	0.25	0	0	0.75
z	0.143	0.571	0.143	0.143

Emission probabilities:

	a	b	c	d
x	0.429	0.286	0.286	0
y	0.6	0	0.2	0.2
z	0.167	0.5	0.167	0.167

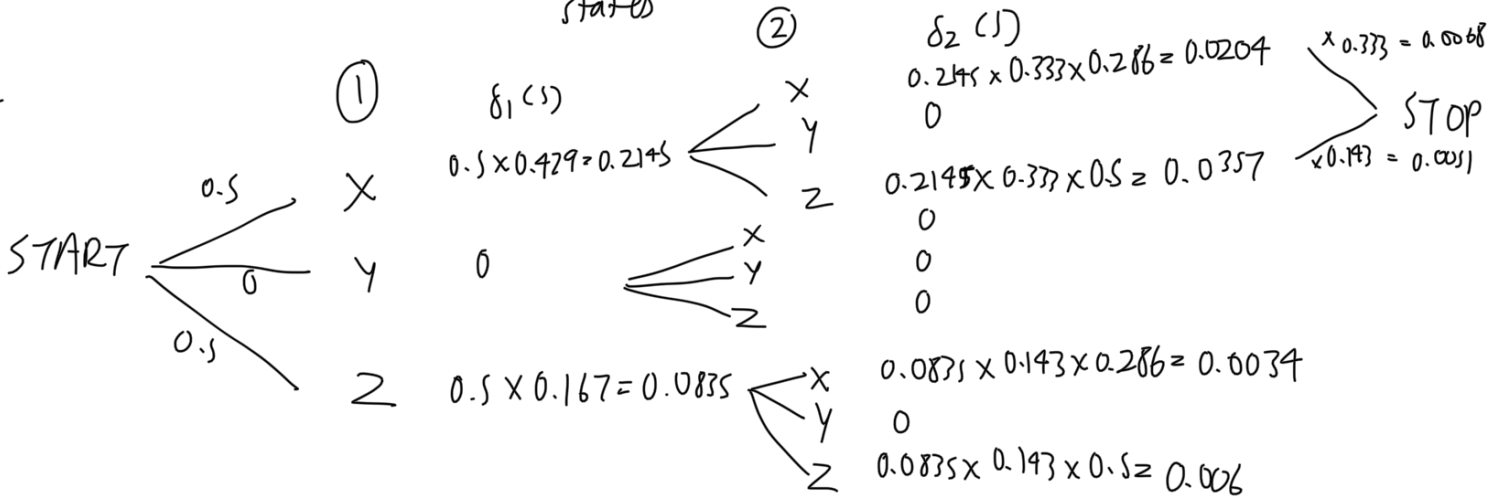
b) Add 1 to every transition count to prevent any 0s.

$$\hat{a}_{u,v} = \frac{\text{count}(u \rightarrow v) + 1}{\text{count}(u) + N}, \quad \hat{b}_{u,v} = \frac{\text{count}(u, v) + 1}{\text{count}(u) + V}$$

where $N = \text{no. of possible next states}$

where $V = \text{no. of observation symbols}$

2.



Most probable state seq. = START $\rightarrow$ x $\rightarrow$ x $\rightarrow$ STOP

3. a) At $i=1$,

$$\delta_1(\text{START}, v) = \pi(v) \cdot b_v(x_1)$$

At $i=2$,

$$\delta_2(u, v) = \delta_1(\text{START}, u) \cdot p(v | \text{START}, u) \cdot b_v(x_2)$$

$$\delta_i(u, v) = \max P(y_i^{i-2}, y_{i-1} = u, y_i = v, x_i)$$

Since transitions depend on prev. 2 states, dynamic prog. table has to track the score for each possible pair of consecutive states.

b) When $i \geq 3$,

$$\delta_i(u, v) = \max \delta_{i-1}(w, u) \cdot p(v | w, u) \cdot b_v(x_i)$$

$$c) \max \delta_n(u, v) \cdot p(\text{STOP} | u, v)$$

Backtracking can be done using $\langle u^*, v^* \rangle$, the final state pair that gives $\max \delta_n(u, v) \cdot p(\text{STOP} | u, v)$.

$$4. \text{ Transition} = P(Z_i | Z_{i-1}) \quad \text{Emission} = P(X_i | Z_i), P(Y_i | Z_i, X_i)$$

Forward: Let $\alpha_i(z) = P(X_i, Y_i, Z_i = z)$

$$\text{Initialization } \alpha_1(z) = P(Z_1 = z | \text{START}) \cdot P(X_1 | z) \cdot P(Y_1 | z, X_1)$$

$$\text{Recursion } \alpha_i(z) = \sum \alpha_{i-1}(z') \cdot P(z | z') \cdot P(X_i | z) \cdot P(Y_i | z, X_i)$$

$$\text{Termination} = \sum \alpha_n(z) \cdot P(\text{STOP} | z)$$

Backward:

$$\text{Let } \beta_i(z) = P(X_{i+1}, Y_{i+1} | Z_i = z)$$

$$\text{Initialization } \beta_n(z) = P(\text{STOP} | z)$$

$$\text{Recursion } \beta_i(z) = \sum P(z' | z) \cdot P(X_{i+1} | z') \cdot P(Y_{i+1} | z', X_{i+1}) \cdot \beta_{i+1}(z')$$

when ^{req.} length = n , no. of states = S ,

$$\text{Time complexity} = O(n \times S \times S) = O(n \cdot S^2)$$